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Genome-Wide Association Study of the Metabolic Syndrome in UK Biobank

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Abstract

Background: The metabolic syndrome (MetS) is a description of a clustering of cardiometabolic risk factors in the same individual. Previous genome-wide association studies (GWASs) have identified 29 independent genetic loci linked to MetS as a binary trait. This study used data from UK biobank to search for additional loci.
Methods: Using data from 291,107 individuals in the UK biobank, a GWAS was performed versus the binary trait MetS (harmonized NCEP criteria).
Results: In a GWAS of MetS (binary) we found 93 independent loci with $P < 5 \times 10^{-8}$, of which 80 were not identified in previous GWASs of MetS. However, the majority of those loci have previously been associated with one or more of the five MetS components. Of particular interest are the genes being related to MetS (binary) in this study, but not to any of the MetS components in past studies, such as *WDR48, KLF14, NAADL1, GADD45G,* and *OR5R1,* as well as the two loci that have been associated with all five MetS components in past studies, *SNX10* and *C5orf67.* A pathway analysis of the 93 independent loci showed the immune system, transportation of small molecules, and metabolism to be enriched.
Conclusion: This GWAS of the MetS in UK biobank identified several new loci being associated with MetS. Most of those have previously been found to be associated with different components of MetS, but several loci were found not previously linked to cardiometabolic disease.

Keywords: metabolic syndrome, gene, UK biobank, GWAS, genetics

Introduction

In the late 1980s, the phenomenon that cardiovascular risk factors tend to cluster in certain individuals was recognized by several groups. This was initially called ''syndrome X,''[1] or ''the metabolic risk factor syndrome,''[2] but in the recent years the term ''metabolic syndrome'' (MetS) has been used to denote this clustering of risk factors.

Since 2007, a large number of large genome-wide association studies (GWASs) have been conducted to identify genetic loci linked to the risk factors included in the definition of MetS, such as diabetes,[3] hypertension,[4] high-density lipoprotein (HDL)/triglycerides,[5] and fat distribution.[6] Also a number of GWASs have been conducted with MetS as a binary trait,[7–10] and 29 independent loci have been identified. One striking feature of these MetS-GWASs is that loci linked to lipid traits are the most prominent findings.

Today the UK biobank provides a unique resource for GWASs, and since the sample size is more than a 10-fold larger than the GWASs conducted for MetS so far, the aim of this study was to identify novel loci for MetS, and to investigate if those loci previously have been reported to be associated with the risk factors included in the definition of MetS.

Methods

Sample

UK Biobank is a large prospective cohort study conducted multicenter across UK (https://www.ukbiobank.ac.uk). In 2006–2010, >500,000 individuals aged 40–69 years underwent physical measurements, and blood samples were biobanked for later analysis of genes and biomarkers. The study was approved by the North West Multi-Centre Research Ethics Committee. All participants provided written informed consent.

Genetic analysis

The genome-wide genetic data were available for 488,377 UK Biobank individuals. Genotyping was performed

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with the UK BiLEVE and UK Biobank Axiom arrays in 488, 377 individuals and imputed with IMPUTE2 by using both HRC and 1000 Genomes Phase 3 merged with the UK10K haplotype reference panels. Quality control of genetic markers consisted of tests for batch effects, plate effects, departures from Hardy–Weinberg equilibrium, gender effects, array effects, and discordance.[11]

The March 2018 release of the imputed genetic marker data (including corrected imputations from the UK10K and 1000 Genomes Phase 3 reference panel) was used, and the sample was restricted to unrelated individuals with self-reported British descent and European ethnicity based on principal component analysis. Furthermore, we excluded genetic markers with minor allele count ≥5822 and imputation quality <0.8. For chromosomal positions, hg19 was used.

The study was conducted in 291,107 individuals with nonmissing data on outcomes, covariates, and genotype data.

Clinical and biochemical data

Serum glucose, HDL-cholesterol and triglycerides were measured by a Beckman coulter AU5800, by standard methods. Since samples were taken with different times of fasting, glucose levels were adjusted down by 1.5 mmol/L if reported fasting time was 0 hr, −3.0 mmol/L if fasting was 1 hr, −1.0 mmol/L if fasting was 2 hr, −0.3 mmol/L if fasting was 3 hr, and no correction if fasting time was >3 hr. For triglycerides, the levels were adjusted down by 0.1 mmol/L if reported fasting time was 1 hr, and the reductions were −0.2, −0.4, −0.6, −0.65, −0.4, and −0.1 mmol/L for times 2–7 hr, respectively. The magnitudes of these adjustments were based on a literature search and own experience of mixed meal studies. Blood pressure was measured twice in the sitting position with the automated Omron device.

The MetS

The harmonized NCEP-criteria for MetS were used to define the five components of the syndrome and prevalent MetS (binary).[12] Three of the following five criteria should be fulfilled: Blood pressure ≥130/85 mmHg or antihypertensive treatment, serum glucose ≥6.1 mmol/L or antidiabetic treatment, serum triglycerides ≥1.7 mmol/L, waist circumference >102 cm in men and >88 cm in women, HDL-cholesterol <1.0 mmol/L in men and <1.3 mmol/L in women.

Statistical methods

A GWAS was performed using 9,463,307 genetic variants versus MetS as a binary trait in logistic regression analysis adjusting for age, gender, genetic analysis batch, and 20 principal components for the population genetics. In this analysis, $P < 5 \times 10^{-8}$ was regarded as significant, and PLINK 2.0 was used for the analysis.

Using Phenoscanner homepage (www.phenoscanner.medschl.cam.ac.uk), a search was performed on previously published GWAS findings for different traits using the name of the nearest gene given in Table 2 for all 93 identified loci. If the gene name was not identified by Phenoscanner, the rs-number was used instead.

Using the Reactome homepage (https://reactome.org, version 68), the names of the nearest gene given in Table 2 for all 93 identified loci were uploaded and a pathway analysis was performed based on gene enrichment. To evaluate enrichment, a binomial test was used to calculate the probability shown for each result.

Results

Basic characteristics of the sample are given in Table 1.

In the GWAS of MetS (binary) we found 93 independent loci with $P < 5 \times 10^{-8}$ (see Table 2 for details, and Manhattan-plot and QQ-plot in Fig. 1). The top three hits were in or close to well-known lipid-regulating genes (the APOA1/APOC3/APOA4/APOA5-cluster and ZPR1, CETP, and LPL). When compared with the previously published GWAS loci for MetS (as binary trait, Supplementary Table S1), 80 of the 93 loci were novel findings. The results from the search of previous GWAS findings for the 93 loci using Phenoscanner are given in detail in Supplementary Table S2.

The findings regarding the different MetS components are given in Table 2. In general, almost all loci have previously been associated with at least one MetS component in previous GWASs, but *WDR48, KLF14, NAADL1, GADD45G,* and *OR5R1* have not previously been linked to cardiometabolic traits. Of interest are also two loci that have been associated with all five MetS components in past studies, *SNX10* and *C5orf67.*

Results from the pathway/gene enrichment analysis are shown in Supplementary Figs. 1 and 2. The most enriched pathways were found in the immune system, followed by transportation of molecules and lipid metabolism.

Discussion

This study performed a whole-genome analysis and found 93 significant loci being related to MetS in UK biobank. The majority of those loci were novel for MetS as a binary trait, but were known loci for the five cardiometabolic components included in MetS.

Comparison with previous GWASs of MetS

Several GWAS of MetS have previously been performed in different populations.[7–10] An analysis of those in Phenoscanner showed 29 independent loci with $P < 5 \times 10^{-8}$,

Table 1. Basic Characteristics of the Sample ($N = 291,107$)

Variables	*Mean (SD) or proportion*
Age	56.9 (7.9)
Gender (% female)	47
Systolic blood pressure (mmHg)	140 (19)
Diastolic blood pressure (mmHg)	82 (10)
Antihypertensive treatment (%)	20.8
Waist circumference (cm)	90 (13)
BMI (kg/m^2)	27.4 (4.7)
HDL-cholesterol (mmol/L)	1.4 (0.4)
Serum triglycerides (mmol/L)	1.3 (1.0)
Serum glucose (mmol/L)	4.7 (1.4)
Antidiabetic treatment (%)	1.0
Prevalence of MetS (%)	20.5

Means and (SD) or proportions are given.
HDL, high-density lipoprotein; MetS, metabolic syndrome; SD, standard deviation.

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Table 2. Genome-Wide Association Study of Metabolic Syndrome (Binary) Disclosed 93 Independent Loci with $P < 5 \times 10^{-8}$

SNP	*Chr*	*Position*	*Effect allele*	*Other allele*	*EAF*	*Beta*	*SE*	P	*Nearest gene*	*Previous MetS trait*
rs964184	11	116648917	G	C	0.132408	0.252628	0.0091752	6.90e-167	ZPR1	HDL, TG
rs247617	16	56990716	A	C	0.324218	−0.196432	0.0071461	2.43e-166	CETP	HDL, TG
rs3844510	8	19861361	C	A	0.26399	−0.191134	0.0076266	1.30e-138	LPL	HDL, TG
rs72836561	17	41926126	T	C	0.0322648	0.23953	0.0174139	4.74e-43	CD300LG*	HDL
rs2980888	8	126507308	T	C	0.299847	0.0959522	0.0070574	4.23e-42	TRIB1*	TG
rs116843064	19	8429323	A	G	0.0191837	−0.337476	0.0263335	1.34e-37	ANGPTL4*	HDL,TG
rs2281721	1	230297136	C	T	0.38621	0.0820747	0.0066785	1.03e-34	GALNT2	HDL,TG, fat mass
rs483082	19	45416178	T	G	0.238624	0.0899299	0.0075775	1.73e-32	APOC1	HDL, TG, WC, diabetes
rs673548	2	21237544	A	G	0.204049	−0.0977899	0.0082417	1.79e-32	APOB	HDL, TG
rs1535	11	61597972	G	A	0.347774	0.0792125	0.0068083	2.74e-31	FADS2*	HDL, TG, Glucose
rs261290	15	58678720	T	C	0.347343	−0.0779563	0.0069174	1.86e-29	LIPC	HDL, TG
rs56094641	16	53806453	G	A	0.402905	0.0736765	0.0066394	1.30e-28	FTO	WC
rs2138161	2	227095159	T	C	0.351852	−0.0752413	0.0068822	8.03e-28	NEU2*	Diabetes, hypertension, fat mass
rs998584	6	43757896	A	C	0.481524	0.0706852	0.0065509	3.83e-27	VEGFA*	Fat mass
rs28597716	8	19936687	G	A	0.189687	−0.152184	0.0086723	1.04e-24	SLC18A1*	Diabetes
rs12056034	7	72878645	G	A	0.124753	−0.0968365	0.0100916	8.33e-22	BAZ1B*	HDL, TG
rs66922415	18	57848651	G	A	0.234519	0.0678461	0.0076254	5.72e-19	MC4R*	WC
rs2925979	16	81534790	T	C	0.301767	0.0616096	0.007063	2.71e-18	CMIP*	Diabetes, hypertension, HDL
rs7124681	11	47529947	A	C	0.408583	0.055786	0.0066152	3.37e-17	CELF1*	WC, HDL, hypertension,
rs11206374	1	40048009	A	G	0.225766	0.065077	0.0077364	4.04e-17	PABPC4*	WC, HDL, TG, diabetes
NA	20	44553722	T	TA	0.187918	0.0694639	0.0082966	5.64e-17	PCIF1*	HDL, TG
rs61789601	3	135954979	T	C	0.202909	−0.0683717	0.008209	8.16e-17	PCCB*	WC, HDL, TG
rs11429307	5	55857025	GT	G	0.190881	0.0669661	0.0082284	4.01e-16	C5orf67*	WC, HDL, TG, diabetes, hypertension
rs11754773	6	34577257	G	A	0.0942279	0.089294	0.0110053	4.91e-16	C6orf106*	WC
rs1260326	2	27730940	T	C	0.392938	0.0536971	0.0066555	7.14e-16	GCKR	TG, glucose
rs139974673	15	44027885	C	T	0.0251253	0.160596	0.0200473	1.14e-15	CATSPER2P1*	HDL, TG, BMI
rs9987289	8	9183358	A	G	0.0919748	0.0873883	0.0110761	3.03e-15	LOC157273	HDL, glucose
rs1800961	20	43042364	T	C	0.0314816	0.138848	0.0180355	1.38e-14	HNF4A*	HDL, diabetes
rs10822155	10	65071215	A	C	0.41581	−0.0511185	0.0066509	1.52e-14	JMJD1C*	TG, blood pressure
rs12599637	16	69593355	C	T	0.403682	−0.0509657	0.0067725	5.26e-14	MIR1538/NFAT5*	—
rs638714	1	62906489	T	G	0.347785	−0.0514954	0.0069156	9.60e-14	USP1*	TG
rs632057	6	139834012	T	G	0.372799	0.0486791	0.0067304	4.73e-13	LINC01625*	TG, HDL, HbA1c, trunk fat
rs10260148	7	130430969	T	C	0.27901	0.0521287	0.0072552	6.72e-13	KLF14*	0
rs114165349	1	27021913	C	G	0.0235174	0.147253	0.0208744	1.74e-12	ARID1A*	TG, HDL, hypertension, trunk fat
rs10187501	2	165532454	G	A	0.346097	−0.0465591	0.0069069	1.57e-11	COBLL1*	TG, HDL, diabetes, WC
rs1532127	19	47571938	G	A	0.314222	−0.0472097	0.0070678	2.40e-11	ZC3H4*	Diabetes, WC
rs7239575	18	21120035	C	T	0.49327	−0.043234	0.0065326	3.64e-11	NPC1*	WC
rs10049088	3	156797648	T	C	0.386891	−0.0443217	0.0067433	4.94e-11	LINC00880*	WC
rs1009360	2	65276049	C	T	0.415241	−0.0435887	0.0066372	5.12e-11	CEP68*	TG, blood pressure
rs7660883	4	87982876	G	C	0.377013	−0.0443878	0.0067925	6.37e-11	AFF1*	TG, HDL
rs11075253	16	15148646	A	C	0.297121	−0.0468788	0.0071929	7.15e-11	PDXDC1*	WC, TG
rs1534696	7	26397239	C	A	0.460436	0.0424362	0.0065434	8.85e-11	SNX10*	TG, HDL, diabetes, WC, hypertension

(*continued*)

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TABLE 2. Continued

SNP	*Chr*	*Position*	*Effect allele*	*Other allele*	*EAF*	*Beta*	*SE*	P	*Nearest gene*	*Previous MetS trait*
rs7563362	2	620297	A	G	0.142333	−0.0613566	0.0095051	1.08e-10	TMEM18*	WC
rs9378248	6	31326289	A	G	0.339668	0.0440684	0.0068853	1.55e-10	HLA-B	Diabetes, trunk fat mass
rs563296	10	99772404	G	A	0.44029	−0.0419399	0.0065928	2.00e-10	CRTAC1*	WC, diabetes
rs1023193	15	41855736	T	G	0.305574	−0.045397	0.0071595	2.29e-10	TYRO3*	BMI, blood pressure
rs13107325	4	103188709	T	C	0.0748556	0.0760904	0.0121877	4.29e-10	SLC39A8*	HDL.WC, blood pressure
NA	12	124503803	CAA	C	0.436617	−0.0414889	0.0066476	4.34e-10	ZNF664*	HDL.WC,TG
rs779518917	17	76398124	GTGT	G	0.403203	0.0415352	0.0066831	5.13e-10	PGS1*	HDL, blood pressure
rs35661464	11	64828842	T	C	0.252033	0.0463799	0.0074696	5.33e-10	NAALADL1*	0
rs764311894	20	51204733	CTTT	C	0.169241	−0.0569809	0.0091796	5.39e-10	LINC01524*	—
rs748326686	4	67851522	CT	C	0.37669	0.042439	0.0068632	6.27e-10	LOC101927237*	—
rs577721086	6	127440047	C	T	0.0509599	0.0900937	0.0147076	9.03e-10	RSPO3*	HDL.WC,TG
rs537069642	3	50119387	CTT	C	0.423828	−0.0418519	0.0068962	1.29e-09	RBM6*	HDL, diabetes, WC, hypertension
rs76376137	6	34173330	G	T	0.050549	0.0820329	0.0146096	1.30e-09	HMGA1*	WC
rs10913469	1	177913519	C	T	0.207314	0.0482641	0.0079955	1.58e-09	SEC16B*	WC
rs56133711	11	27723334	A	G	0.262839	0.0443302	0.0073785	1.88e-09	BDNF*	WC
rs3814883	16	29994922	T	C	0.48348	0.0392918	0.0065507	2.00e-09	TAOK2*	WC, hypertension
rs56282717	7	150657095	A	G	0.244425	−0.0460964	0.0076943	2.09e-09	KCNH2	WC, hypertension
rs12945575	17	40713071	T	C	0.25004	0.044779	0.0075111	2.50e-09	COASY*	WC
rs2306363	11	65405600	T	G	0.206589	−0.0482761	0.0081366	2.97e-09	SIPA1*	Hypertension
rs781006834	7	17941865	G	GT	0.344659	0.0411062	0.0069362	3.10e-09	SNX13*	HDL
rs11751347	6	161092438	T	C	0.102714	0.0624875	0.0106566	4.53e-09	LPA*	WC, TG
rs73123462	4	36077604	T	C	0.0155967	0.149549	0.0255208	4.63e-09	ARAP2*	Diabetes
rs62107261	2	422144	C	T	0.0482567	−0.089776	0.0156627	5.05e-09	LINC01874*	WC
rs3808439	8	116563675	A	G	0.447923	0.0383457	0.0065731	5.42e-09	TRPS1*	WC, HDL
rs56959712	12	123188475	T	G	0.210819	−0.0470487	0.0080746	5.65e-09	HCAR2*	HDL, fat mass
rs56001710	7	25983400	A	T	0.417154	0.0388139	0.0068186	6.85e-09	MIR148A*	—
rs779470261	3	131646163	C	CTCTAA	0.26329	0.0431103	0.0074441	6.99e-09	CPNE4*	WC
rs5789783	11	13347748	T	TA	0.402641	−0.0399511	0.0069148	7.57e-09	ARNTL*	WC, hypertension
rs3949781	9	92178472	T	A	0.461633	−0.0380199	0.0065968	8.25e-09	GADD45G*	0
rs11871285	17	65840809	T	G	0.193205	0.0473409	0.0082208	8.48e-09	BPTF*	WC
rs6545703	2	59032635	G	T	0.415927	−0.0382425	0.0066477	8.78e-09	LINC01122*	WC
rs58175144	1	150839698	CA	C	0.301172	0.0413753	0.0072197	9.99e-09	ARNT*	HDL, trunk fat mass
rs7188873	16	24727064	A	G	0.376011	−0.03863	0.0067729	1.17e-08	TNRC6A*	WC, blood pressure
rs5021727	6	32578633	G	A	0.461566	0.0371336	0.0065596	1.51e-08	HLA-DRB1*	Diabetes
rs1105654	18	47147746	G	A	0.345091	0.038799	0.0068934	1.82e-08	LIPG*	HDL
rs12752223	1	93837133	T	C	0.396475	−0.0374831	0.0066755	1.97e-08	DR1*	HDL
rs140628616	11	56201709	A	AAT	0.16585	−0.0528712	0.0094553	2.25e-08	OR5R1*	0
rs9971210	10	21879084	G	C	0.489597	0.0365804	0.0065457	2.29e-08	MLLT10*	WC, blood pressure
rs10954772	8	30863938	T	C	0.313991	0.0391984	0.0070493	2.69e-08	PURG*	WC, hypertension
rs12472667	2	171629063	G	C	0.371305	0.0374895	0.0067634	2.97e-08	ERICH2*	WC
rs1143015	17	7485206	A	G	0.157971	0.0491401	0.0088808	3.14e-08	CD68*	Weight
rs11789603	9	107647019	T	C	0.107873	−0.058891	0.0106912	3.62e-08	ABCA1	HDL

(*continued*)

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TABLE 2. Continued

SNP	*Chr*	*Position*	*Effect allele*	*Other allele*	*EAF*	*Beta*	*SE*	P	*Nearest gene*	*Previous MetS trait*
rs9332817	11	118365210	C	G	0.0265006	−0.115795	0.0210291	3.66e-08	KMT2A*	WC
rs535286942	10	94797201	AT	A	0.374794	−0.037682	0.0068466	3.72e-08	EXOC6*	Diabetes
rs8121509	20	62712053	C	T	0.451827	−0.0360691	0.0065647	3.92e-08	OPRL1*	WC, hypertension
rs143321598	3	39116681	CT	C	0.488259	−0.0382859	0.0069834	4.20e-08	WDR48*	0
rs10945840	6	162974904	C	G	0.332614	−0.0382931	0.0069905	4.30e-08	PARK2*	—
rs7138803	12	50247468	A	G	0.368675	0.0369574	0.0067465	4.30e-08	BCDIN3D*	WC
rs11655056	17	47364107	C	T	0.46087	0.0359929	0.0065729	4.35e-08	MIR6129*	0
rs4921913	8	18272377	C	T	0.221605	0.0426441	0.0077967	4.51e-08	NAT2*	TG
rs2307111	5	75003678	C	T	0.393101	−0.0366715	0.006708	4.58e-08	POC5*	WC

The loci given with an *asterisk* after the name of the nearest gene are novel findings for MetS as binary trait. The beta represents the log odds. EAF, effect allele frequency. Build37, hg19 is used for chromosomal positions. Previously published associations with MetS components are given in the right-hand-side column using data from Phenoscanner. Details on those findings are given in Supplementary Table S2—denotes no associations versus any trait. Zero denotes no association with a MetS trait, but associations with other traits.

and of those, 13 could be found in the present GWAS, leaving 80 novel loci for MetS found in this study.

As pointed out in a review of the previous MetS-GWAS by Brown and Walker,[13] the major loci in the previous GWASs were in or close to lipid-regulating genes, such as *APOB, LPL, CETP, APOA5, GCKR,* and *ZNF259*, suggesting lipid metabolism to be a major driver in MetS. Those lipid-regulating genes are also among the top findings in the present GWAS in UK biobank. It is, however, not strange that lipid-regulating genes are in top of the list of GWAS hits of MetS for two different reasons. First, two of the five components of MetS are defined by lipids (HDL and triglycerides), and since there is a large overlap between genes linked to triglycerides and HDL, such genes have an increased chance of receiving low *P* values in a GWAS of MetS. Second, many of the lipid-regulating genes have a larger impact on lipid traits compared with the impact of single genes on the more complex traits, such as blood pressure, glucose, and visceral obesity. Taken together, it could, therefore, not be taken as granted that lipid-regulating genes are the principal drivers of the clustering of cardiometabolic risk factors just because they are among the top GWAS hits for MetS.

Novel MetS loci in relation to MetS components

Although most of the loci found in the present GWAS were novel loci for MetS as a binary trait, most of them are not novel in terms of associations with the risk factors included in MetS. As could be seen in Table 2, all but 11 of the identified genes have previously been linked to MetS components, or closely related traits such as fat mass, in other GWASs. This is a most likely finding, since MetS is a summary of these cardiometabolic traits. Of particular interest are the genes being related to MetS, but not to any of the MetS components, such as *WDR48, KLF14, NAADL1, GADD45G,* and *OR5R1*. It is not possible to find any common pathophysiological theme linking those genes together, but if the associations between those loci and MetS could be reproduced in other studies they might be a great interest as a clue to the pathway(s) responsible for the clustering of cardiometabolic risk factors in certain individuals.

Another interesting observation is that while the combination of previous associations with HDL and triglycerides are seen for the 11 top-ranked loci in the present MetS-GWAS, these traits are not very common among the genes further down the list. Among those not top-ranked genes, a previous known association with waist circumference is a dominant finding, emphasizing the important role of abdominal fat distribution for MetS.

A third observation is that only two loci have been reported to be linked to all five of the MetS components, *SNX10* and *C5orf67*. Sorting Nexin 10 (SNX10) is a protein-coding gene. It is involved in protein sorting and membrane trafficking in endosomes. It is required for osteoclast formation,[14] and a mutation in this gene has been linked to osteopetrosis at young age.[15] This gene has also been linked to coronary heart disease,[16] and to death in stroke in the UK biobank sample. *C5orf67* is another protein-coding gene, but yet of unknown action. Also this loci has been linked to coronary heart disease.[17] Since both of these loci have been linked to all five components of MetS, they are of great interest for further studies in the pathophysiology of clustering of risk factors.

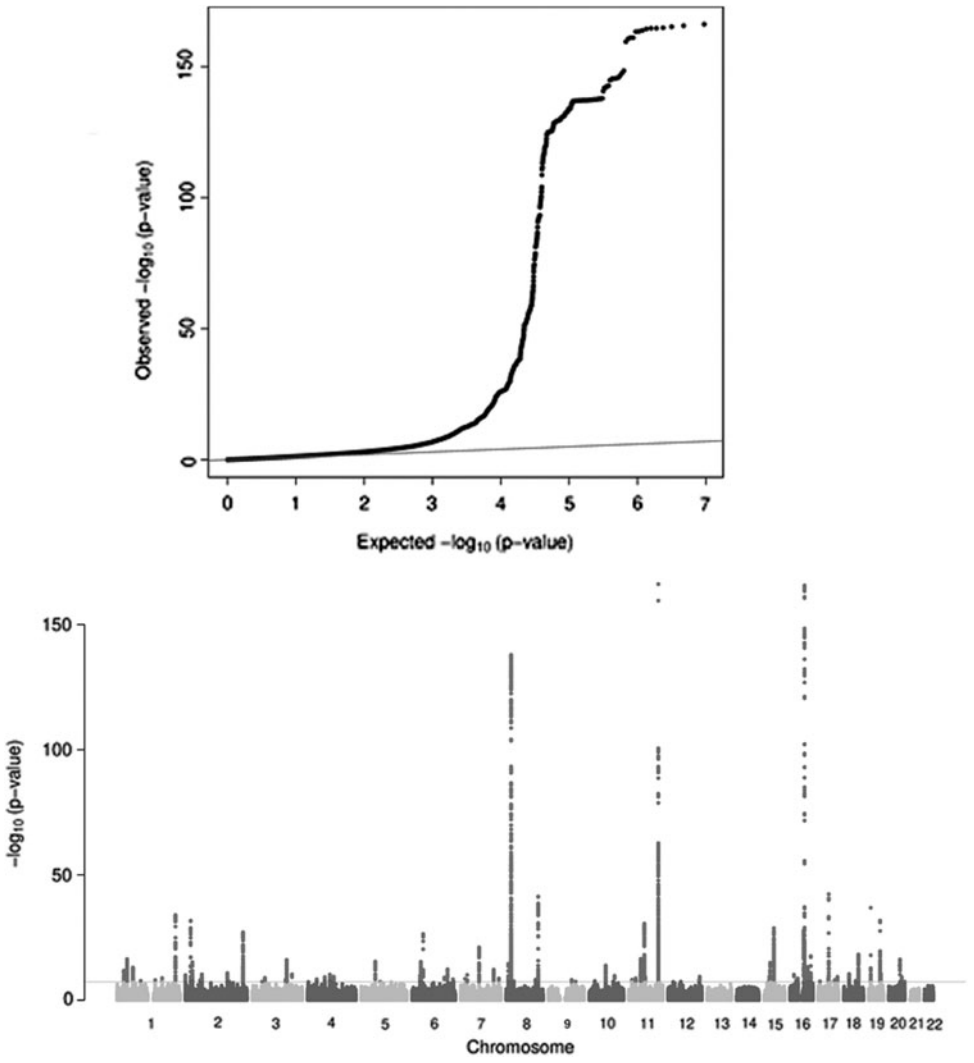


FIG. 1. A Q-Q plot (*upper panel*) and a Manhattan-plot (*lower panel*) for the GWAS for MetS. The Q-Q plot shows the observed *P* value versus the expected *P* value (on a −10 log-scale), whereas the Manhattan-plot shows the observed *P* value (on a −10 log-scale) for each genetic locus in relation to the chromosomal position. GWAS, genome-wide association study; MetS, metabolic syndrome.

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Pathway analysis

A pathway analysis performed on the loci identified in our GWAS showed enrichment in several pathways. To notice is that the majority of those pathways are involved in immunity and inflammation. Also pathways involved in transport of small molecules, such as endosomal/vacuolar and ER-phagosome pathways, and to a lesser degree metabolism, represented by pathways involved in the regulation of lipids and lipoproteins.

The first link described between inflammation and MetS was an association with elevated levels of C-reactive protein.[18–20] These observations were later expanded by associations with other proinflammatory molecules, such as tumor necrosis factor-alpha, interleukin-8, and ICAM-1.[21–23]

Using flow cytometry, it was later shown that subjects with MetS showed increased levels of eosinophils, $CD3^+$ and $CD8^+$ T cells and Mon2/Mon3 monocytes. On the contrary, Mon1 monocytes were decreased[23] 31109070. It has also been found that resident immune subsets are present in adipose tissue in MetS, suggesting that resident T-effector and invariant natural killer T cells could be involved.[24]

In accordance with the present findings that the interferon pathway is identified by gene enrichment, type I interferon (INF-I) responses in the liver in MetS has been described.[25]

Thus, both past experience and the present pathway analysis supports an important role of inflammation and immune regulation in MetS.

Strengths and limitations

The major strength of this study is the large sample size that results in a greater power than previously published GWASs, which improves the chance to find novel genes associated with MetS.

A limitation is that glucose and triglycerides were not measured in the fasting state in all individuals (HDL is not affected by fasting to a major degree). However, we used a correction based on the time of fast to account for that issue. It is, therefore, reassuring that the major lipid genes found in other GWASs was also found in this study and that several of the loci also previously have been linked to diabetes. Thus, misclassification due to the different time of fasting does not seems to be a major issue, and could if anything only drive the hypothesis testing toward null, and if anything, only produce false negative findings.

Another limitation is that we do not have any large independent sample for replication of our findings, since previous GWASs have been rather small and no other large sample have performed GWASs of MetS. We have, therefore, used

databases such as Phenoscanner to search for validation of our major findings. However, still our findings need replication in another large sample to be conclusive.

In conclusion, this GWAS of the MetS in UK biobank identified several new loci being associated with MetS. Most of those have previously been found to be associated with different components of MetS, but several loci were found not previously linked to cardiometabolic disease.

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Author Disclosure Statement

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Supplementary Material

Supplementary Figure S1
Supplementary Figure S2
Supplementary Table S1
Supplementary Table S2

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